INSTITUT FOR MATEMATIK
AARHUS UNIVERSITET

Ute Hahn
November 14, 2025

Statistical Models

Week 12: Linear models, glms, and "penalized" regression in a Bayesian setting

This week's main topic is the practical analysis of linear models in a Bayesian way, starting with normal distributed models.

For normal linear regression models, we will notice interesting facts about priors: some priors have "shrinkage effects" which means that they implicitely perform variable selection. To understand this, we will shortly look at ridge regression and LASSO in the frequentist world —some of you might know these methods from other courses.

Finally, we will discuss Bayesian analysis of generalized linear models, conveniently using the JAGS machinery. For binary data, we can set up a particular Gibbs algorithm that incorporates a latent variable — you might recall this type of modelling from the first part of the course.

Relevant Readings

- ☐ Analysis of normal means: Reich & Ghosh, Section 4.1
- ☐ Linear models and variable selection: Reich & Ghosh, Section 4.2
- ☐ Generalized linear models: Reich & Ghosh, Section 4.3

Lecture Material

- Link to onenotes on Google drive
- Toy example ridge and LASSO regression

Exercises

Aufgabe 21 zuletzt, som ugens nød

Problem 12.1: MCMC convergence in an overly complicated model

Reich & Ghosh, Chapter 3, Exercise 17.
Consider the (overly complicated) model

$$Y_i \mid \mu_i, \sigma_i^2 \sim \text{Normal}(\mu_i, \sigma_i^2)$$

where

$$\begin{aligned} \mu_i &\overset{\text{i.i.d.}}{\sim} \text{Normal}(0, \theta_1^{-1}), \\ \sigma_i^2 &\overset{\text{i.i.d.}}{\sim} \text{InvGamma}(\theta_2, \theta_3) \text{ and} \\ \theta_j &\sim \text{Gamma}(\epsilon, \epsilon) \text{ for } j = 1, 2, 3. \end{aligned}$$

Assume that the data are $Y_i = i$ for $i = 1, \dots, n$

a. Describe why convergence may be slow for this model.
b. Fit the model in JAGS with all four combinations of $n \in \{5, 25\}$ and $\epsilon \in \{0.1, 10\}$ and report the effective sample size for θ_1, θ_2 and θ_3 for all four fits.

(c) Comment on the effect of n and ϵ on convergence.

Problem 12.2: Effective sample size

This small exercise extends Exercise 5 from Weekly Notice 10, and asks for your creativity: Explain and illustrate the concept of effective sample size using simulation of AR(1) chains. You could show that for the AR(1) chain

$$X_n = \phi X_{n-1} + \lambda + \tau Z_n, \quad n = 1, 2, \dots,$$
$$-1 < \phi < 1,\ \lambda \in \mathbb{R},\ \tau > 0, \quad Z_1, Z_2, \dots \overset{\text{i.i.d.}}{\sim} N(0,1)$$

the "Time-series ESS" given as equation (3.20) in Reich and Ghosh is equal to

$$ESS = S \cdot \frac{1-\phi}{1+\phi},$$

and / or confirm this result by simulation. You will notice that the variance predicted using this effective sample size holds not well for small values of S, due to the simplifications in the formula. It also depends on how close $|\phi|$ is to zero.
Find code for simulation of AR(1) chains here in the course supplementary.

Problem 12.3: Data analysis: comparison of two groups

Reich & Ghosh, Chapter 4, Exercise 1.
A clinical trial gave six subjects a placebo and six subjects a new weight loss medication. The response variable is the change in weight (pounds) from baseline (so -2.0 means the subject lost 2 pounds). The data for the 12 subjects are:

Placebo	Treatment
2.0	-3.5
-3.1	-1.6
-1.0	-4.6
0.2	-0.9
0.3	-5.1
0.4	0.1

a. Conduct a Bayesian analysis to compare the means of these two groups. Would you say the treatment is effective? Is your conclusion sensitive to the prior?

b. Do a corresponding frequentist analysis for comparison. Give reasons for your model choice. Good to know: the t-test is quite robust against violation of the normality assumption, at least when sample size is not very small.

Problem 12.4: Bayesian LASSO for Boston homes

Reich & Ghosh, Chapter 4, Exercise 2.
Load the classic Boston Housing Data in R:

```
library(MASS)
data(Boston)
?Boston
```

The response variable is `medv`, the median value of owner-occupied homes (in $1, 000s), and the other 13 variables are covariates that describe the neighborhood.

a. Fit a Bayesian linear regression model with uninformative Gaussian priors for the regression coefficients. Verify the MCMC sampler has converged, and summarize the posterior distri-

bution of all regression coefficients.

b. Perform a classic least squares analysis (e.g., using the `lm` function in R). Compare the results numerically and conceptually with the Bayesian results.

c. Refit the Bayesian model with double exponential priors for the regression coefficients, and discuss how the results differ from the analysis with uninformative priors.

d. Fit a Bayesian linear regression model in (a) using only the first 500 observations and compute the posterior predictive distribution for the final 6 observations. Plot the posterior predictive distribution versus the actual value for these 6 observations and comment on whether the predictions are reasonable.

Before you do the analysis, you may want to read the documentation of the dataset, and speculate what influence the variables will have on home prices.

Problem 12.5: US gun control and homicides: Poisson and Negative Binomial regression

Reich & Ghosh, Chapter 4, Exercise 4.

☞To understand this exercise, please read the data description on page 155, R&G.
Download the US gun control data here or on the `data` folder on Google Drive, see more in the hint below. These data are taken from the cross-sectional study by Kalesan et al (2016) (reference [47] in R&G).
For state i, let Y_i be the number of homicides and N_i be the population.

a. Fit the model $Y_i \mid \beta \sim \text{Poisson}(N_i\lambda_i)$ where $\log(\lambda_i) = X_i\beta$. Use uninformative priors and $p = 7$ covariates in X_i: the intecept, the five confounders Z_i, and the total number of gun laws in state i. Provide justification that the MCMC sampler has converged and sufficiently explored the posterior distribution, and summarize the posterior of β. Also discuss why the five variables to be included in the analysis are potential confounders.

b. Fit a negative binomial regression model and compare with the results from Poisson regression.

c. Refit the Bayesian model with double exponential priors for the regression coefficients, and discuss how the results differ from the analysis with uninformative priors.

d. Fit a Bayesian linear regression model in (a) using only the first 500 observations and compute the posterior predictive distribution for the final 6 observations. Plot the posterior predictive distribution versus the actual value for these 6 observations and comment on whether the predictions are reasonable.

Hints

When you download the data, you will get a file `guns.RData`. To include the information in your R environment, run `load("guns.RData")`, or in RStudio, just click on the file. You will then get vectors `Y` and `N`, and matrices `Z` and `X`. Matrix `X` containes indicator variables for each of the 25 gun laws per state, so you have to calculate the total number of laws for each state yourself :-).
JAGS hint: since λ_i in the model stands for a rate, you would need to multiply it with the number N_i in the likelihood part of the model specification code (but not in the link function). If you use Listing 4.7, replace `lambda[i]` by `N[i]*lambda[i]` in lines 18 and 26.

I recommend to do at least one of the three data analysis problems 12.3, 12.4 or 12.5.